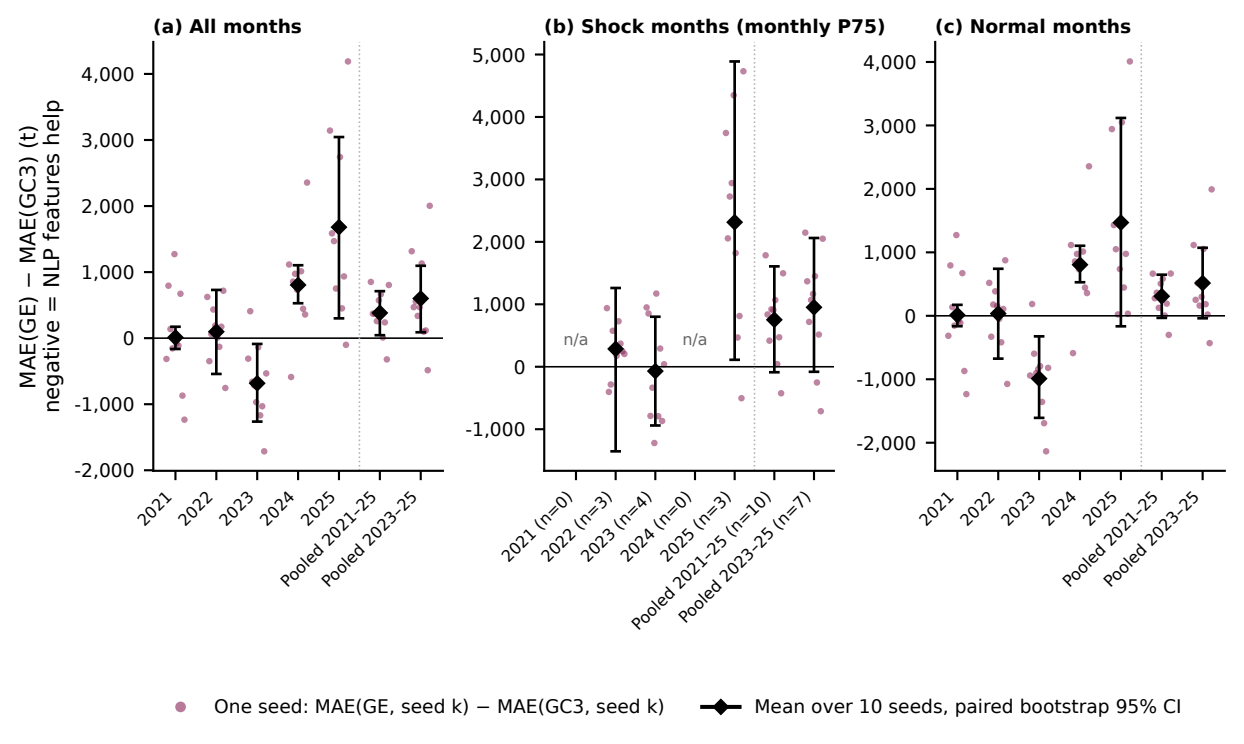
(a) All months
(b) Shock months (monthly P75)
(c) Normal months
MAE(GE) − MAE(GC3) (t)
negative = NLP features help
4,000
3,000
2,000
1,000
0
-1,000
-2,000
5,000
n/a
n/a
2021
2022
2023
2024
2025
Pooled 2021–25
Pooled 2023–25
2021 (n=0)
2022 (n=3)
2023 (n=4)
2024 (n=0)
2025 (n=3)
Pooled 2021–25 (n=10)
Pooled 2023–25 (n=7)
One seed: MAE(GE, seed k) − MAE(GC3, seed k)
Mean over 10 seeds, paired bootstrap 95% CI